

Freight Transportation Planning and Operations

Freight Supply and Capacity Issues

Freight Supply Concept

- Freight supply represents the **available infrastructure and services** that allow goods to move through a transportation system.
It includes:
- **Physical capacity** – roads, rail tracks, ports, terminals, warehouses.
- **Service capacity** – number of vehicles, frequency of trips, load capacity.
- **Operational capacity** – efficiency of terminals, scheduling, traffic management.

Capacity Issues

Type of Capacity Constraint	Description	Example
Infrastructure	Physical limits on transport networks	Port congestion in Mumbai delays ships
Operational	Inefficient scheduling or lack of coordination	Empty truck return trips after delivery
Institutional	Regulatory or policy barriers	Border delays due to customs checks
Environmental	Limits due to sustainability or land use	Restrictions on truck entry in cities during peak hours

Examples

- **Port of Los Angeles** – faces ship congestion due to surges in container traffic and limited berth space.
- **Indian Railways Freight Corridors** – built to increase capacity and segregate freight from passenger traffic.
- **Urban Delivery Limits** – cities like London and Delhi restrict truck access during daytime to manage congestion.

Freight Productivity and Performance

- **Freight Productivity**
- Productivity measures how efficiently freight resources (vehicles, labor, fuel, time) are used to move goods.

Indicator	Definition	Example
Ton-kilometers (ton-km)	Volume × Distance	1 ton moved 100 km = 100 ton-km
Load Factor	Ratio of actual load to vehicle capacity	Truck carrying 8 tons of 10-ton capacity → 80%
Vehicle Utilization	Trips completed per day/week	A truck making 4 trips/day vs 2 trips/day
Turnaround Time	Time between arrival and next departure	Port turnaround = 1 day vs 3 days
Empty Return Rate	% of trips with no load	Lower rate = higher productivity

Performance Measurement

- Performance includes **speed, reliability, cost, and environmental efficiency.**

Dimension	Measure	Example
Time Efficiency	Average travel time	6 hrs for 300 km route
Reliability	Variability in delivery times	±30 mins delivery window
Cost Efficiency	Cost per ton-km	\$0.05/ton-km
Energy & Emissions	Fuel use, CO ₂ per ton-km	Rail = lower emissions than road

Example Comparison

Mode	Average Cost (\$/ton-km)	Emissions (g CO ₂ /ton-km)	Typical Use
Road	0.05–0.15	60–150	Medium- distance, perishable goods
Rail	0.03–0.05	20–40	Bulk commodities
Water	0.02	10–20	International shipments

Distribution of Freight Flows

Distribution of freight flows means how **goods move across space and modes**, reflecting:

- Economic geography (production and consumption centers).
- Transport infrastructure networks.
- Trade routes and logistics hubs.

Determinants of Freight Flows

- **Economic Activity:** Industrial clusters (e.g., Pune automotive hub).
- **Population & Consumption:** High demand areas attract inbound freight.
- **Infrastructure:** Ports, intermodal terminals, and highways guide flow patterns.
- **Policy & Trade Agreements:** e.g., regional corridors like the **Delhi–Mumbai Industrial Corridor (DMIC)**.

Visualization Example

- **Concept: Freight Corridors**
- Dedicated freight routes improve reliability and reduce congestion.
Example:
- **Eastern & Western Dedicated Freight Corridors (India)**
 - Heavy freight trains run at higher speed and capacity.
 - Reduces transit time and cost.

Concept	Focus	Example	Key Metric
Freight Supply & Capacity	Infrastructure and service availability	Port congestion	Throughput (tons/day)
Freight Productivity	Efficiency of moving goods	Load factor, vehicle utilization	Ton-km, cost/ton-km
Freight Flow Distribution	Spatial and modal movement of goods	Industrial-to-port corridors	Ton-km by mode or region

Production / Consumption → Origin / Destination (O–D) Patterns

Freight transport begins with understanding where goods **are produced** and where they **are consumed**.

- **Production zones** generate freight (e.g., farms, factories, mines).
- **Consumption zones** attract freight (e.g., cities, retail centres, export terminals).
- O–D matrices quantify the **volume of goods** moving from each origin to each destination.
- These patterns are fundamental inputs for:
 - Network design
 - Mode choice modelling
 - Route assignment
 - Infrastructure planning

Competing Modes and Commodity-Specific Mode Choice

Shippers choose between modes based on commodity characteristics and logistics requirements.

Common Modes

- Road (trucks)
- Rail
- Waterway / maritime
- Air
- Pipeline

Commodity Factors Influencing Mode Choice

- **Value per tonne:** high-value goods → air; low-value bulk → rail/water.
- **Urgency / perishability:** perishable goods → air/road.
- **Volume & weight:** heavy volumes → rail/water/pipeline.
- **Handling needs:** hazardous materials may require specific modes.

Cost & Performance Factors

- Transport cost (line-haul + terminal handling)
- Transit time & reliability
- Service frequency
- Safety & risk
- Accessibility (door-to-door vs hub-and-spoke)

Shippers often use **multimodal** combinations to balance cost and speed.

Route Planning

Determines the **paths** freight takes through the transport network.

Key components

- Network characteristics (capacity, congestion, road class, weight limits)
- Cost functions (distance, time, tolls, fuel, reliability)
- Constraints (bridge heights, hazmat restrictions, seasonal closures)

Tools and methods

- Shortest-path algorithms (Dijkstra, A*, etc.)
- GIS-based routing
- Vehicle Routing Problem (VRP) variants for multi-stop logistics
- Dynamic routing using real-time data

Scheduling

Scheduling ensures resources (vehicles, drivers, loading docks) are used efficiently.

Includes:

- Departure/arrival times
- Driver hours of service
- Delivery windows (time-window constraints)
- Fleet assignment and rotation
- Terminal operations (loading, unloading, sequencing)

Good scheduling reduces:

- Idle time
- Late deliveries
- Operating cost
- Congestion impacts

Collection, Storage, and Distribution Centres

Freight systems rely heavily on logistics facilities:

Collection

- First-mile pickup from producers or suppliers
- Consolidation at local depots
- Use of small vehicles for flexible access

Storage / Warehousing

- Buffer between production and consumption
- Inventory management (FIFO, JIT, safety stock)
- Cross-docking to minimize dwell time
- Cold-storage or specialized facilities when required

Distribution Centres (DCs)

- Break-bulk, consolidation, and transshipment
- Located strategically to minimize total logistics cost
- Often near highways, rail terminals, or ports
- Automation, sorting systems, robotics increasingly used

Regulation and Enforcement of Freight Transport

Governments regulate freight to ensure safety, reduce environmental impacts, protect infrastructure, and maintain fair competition.

Key Areas

- **Vehicle regulations:** weight limits, axle loads, dimensions, emissions standards
- **Driver regulations:** licensing, hours of service, fatigue rules
- **Safety regulations:** hazmat handling, vehicle inspections
- **Infrastructure use regulations:** tolling, road pricing, restricted zones
- **Maritime/aviation regulations:** IMO, ICAO standards
- **Enforcement methods:** weigh stations, roadside inspections, digital tachographs, electronic logging devices (ELDs)

Input Category

Freight Load

Truck Capacity

Distance

Fuel Efficiency

Fuel Price

Driver Wage

Average Speed

Output Category

Delivered Freight

Delivery Time

Total Fuel Used

Total Cost

Resource Utilization

Description

Total goods to be transported

Carrying capacity per truck

A → B

Truck mileage

Cost per liter

Per trip

Truck speed

Numerical Output

20 tons

5 hours

140 liters

\$468

2 trucks used

Numerical Value

30 tons

12 tons per truck

400 km

6 km per liter

\$1.30 per liter

\$180 per truck

80 km/h

Explanation

All goods successfully transported

Time from A to B

Fuel needed for all trucks

Operational cost for the trip

Full capacity per truck